

Africast-Time Series Analysis & Forecasting Using R

<https://workshop.f4sg.org/africast/>

2. Time series patterns and basic graphics



Outline

- 1 Time series Patterns
- 2 Time plots
- 3 Seasonal plots
- 4 Seasonal or cyclic?
- 5 Lag plots and autocorrelation
- 6 White noise

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Key patterns of time series

- Level
- Underlying trend
- Seasonal/cycle
- Autocorrelation
- Unpredictable patterns/Noise
- Different types of events and driving factors (i.e. predictors) may affect the time series

Time series patterns

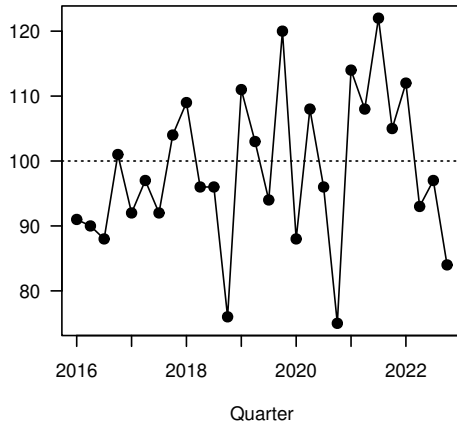
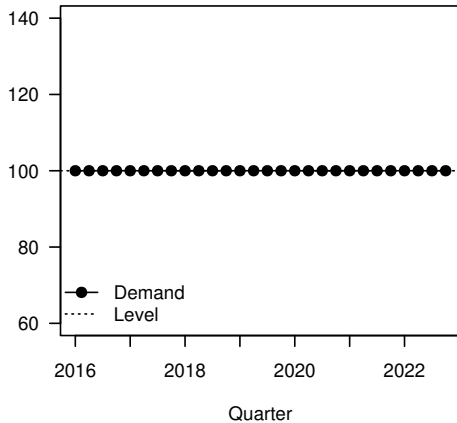
Level The *level* of a time series describes the center of the series.

Trend A *trend* describes predictable increases or decreases in the level of a series.

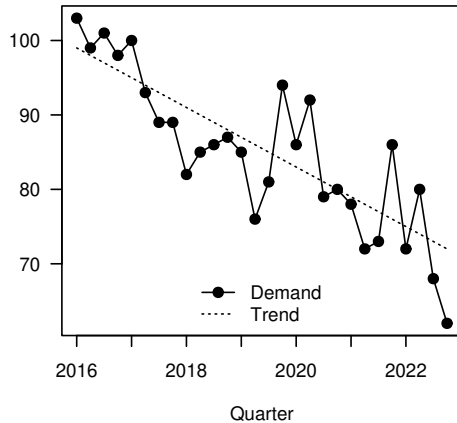
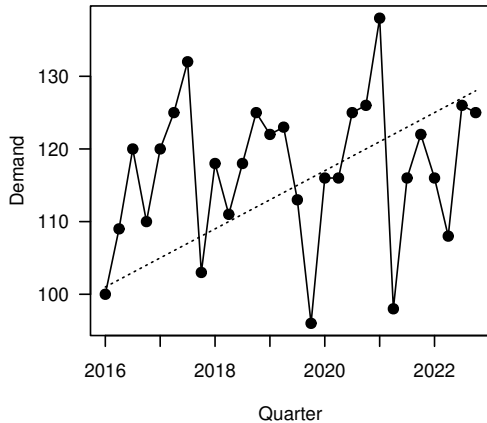
Seasonal *Seasonality* is a consistent pattern that repeats over a fixed period of time. pattern exists when a series is influenced by seasonal factors (e.g., the quarter of the year, the month, or day of the week).

Cyclic pattern exists when data exhibit rises and falls that are *not of fixed period* (duration usually of at least 2 years).

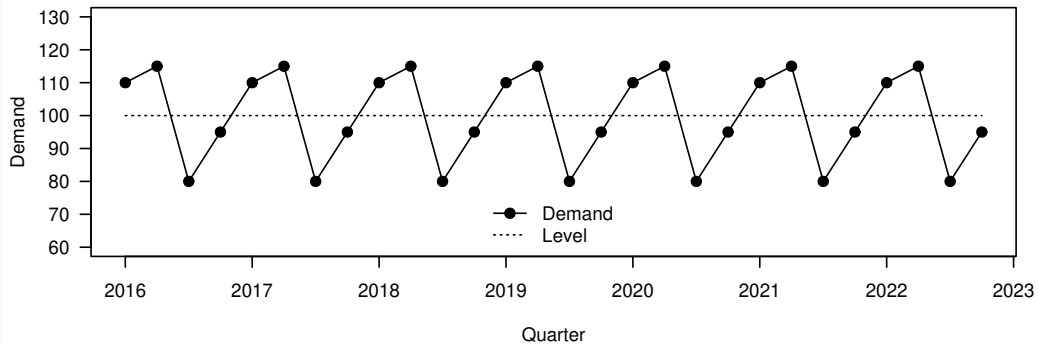
Level



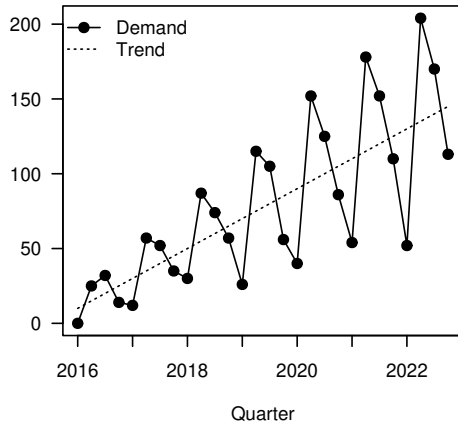
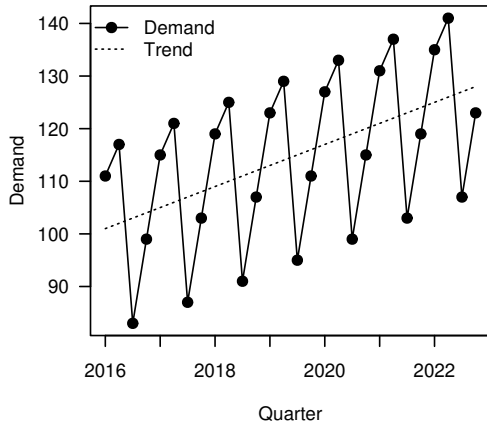
Trend



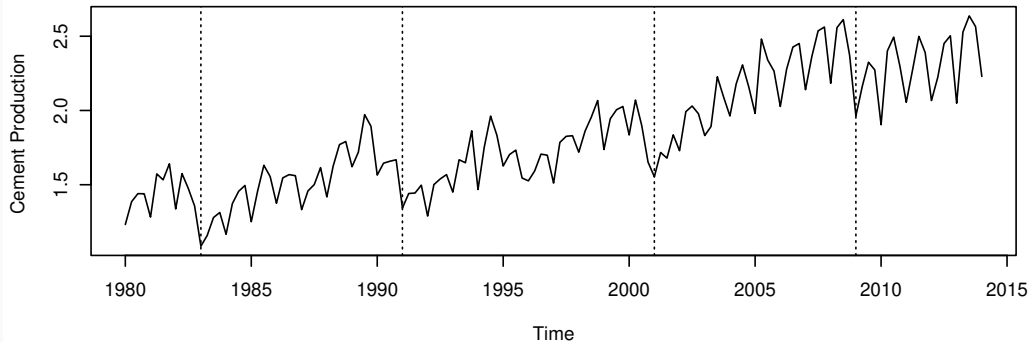
Seasonality



Additive versus multiplicative seasonality



Cycles

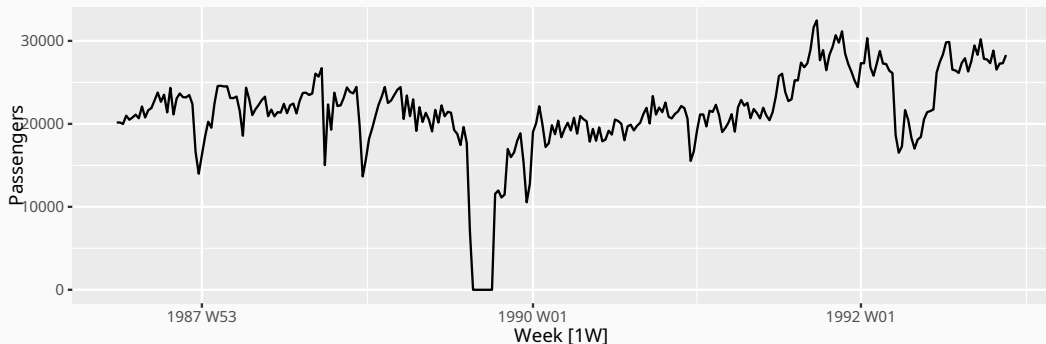


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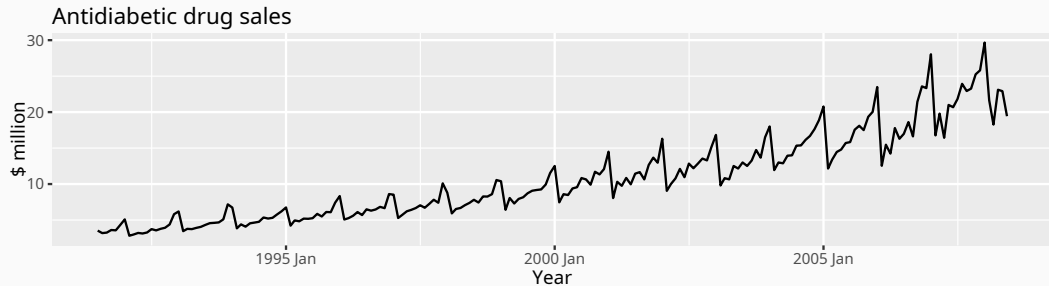
Time plots

```
ansett %>%  
  filter(Airports=="MEL-SYD", Class=="Economy") %>%  
  autoplot(Passengers)
```



Time plots

```
PBS %>% filter(ATC2 == "A10") %>%  
  summarise(Cost = sum(Cost)/1e6) %>% autoplot(Cost) +  
  ylab("$ million") + xlab("Year") +  
  ggtitle("Antidiabetic drug sales")
```



Outline

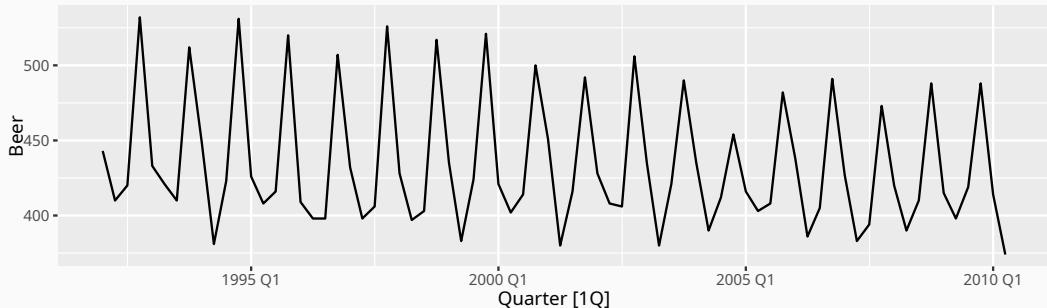
- 1 Time series Patterns
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Seasonal plots

- Data plotted against the individual “seasons” in which the data were observed. (In this case a “season” is a month.)
- Something like a time plot except that the data from each season are overlapped.
- Enables the underlying seasonal pattern to be seen more clearly, and also allows any substantial departures from the seasonal pattern to be easily identified.
- In R: `gg_season()` (`ggtime`)

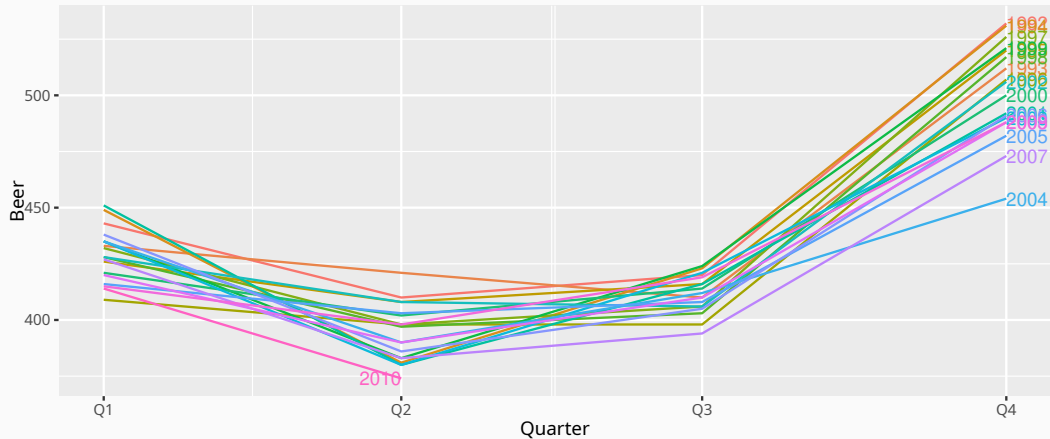
Quarterly Australian Beer Production

```
beer <- aus_production |>  
  select(Quarter, Beer) |>  
  filter(year(Quarter) >= year(1992L))  
beer |> autoplot(Beer)
```



Quarterly Australian Beer Production

```
beer |> gg_season(Beer, labels = "right")
```



Multiple seasonal periods

vic_elec

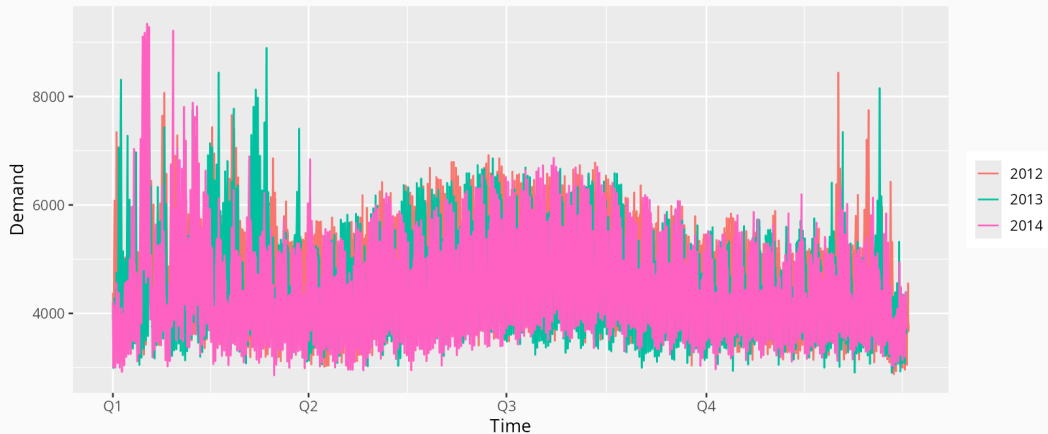
```
# A tsibble: 52,608 x 5 [30m] <Australia/Melbourne>
```

	Time	Demand	Temperature	Date	Holiday
	<dtm>	<dbl>	<dbl>	<date>	<lgl>
1	2012-01-01 00:00:00	4383.	21.4	2012-01-01	TRUE
2	2012-01-01 00:30:00	4263.	21.0	2012-01-01	TRUE
3	2012-01-01 01:00:00	4049.	20.7	2012-01-01	TRUE
4	2012-01-01 01:30:00	3878.	20.6	2012-01-01	TRUE
5	2012-01-01 02:00:00	4036.	20.4	2012-01-01	TRUE
6	2012-01-01 02:30:00	3866.	20.2	2012-01-01	TRUE
7	2012-01-01 03:00:00	3694.	20.1	2012-01-01	TRUE
8	2012-01-01 03:30:00	3562.	19.6	2012-01-01	TRUE
9	2012-01-01 04:00:00	3433.	19.1	2012-01-01	TRUE
10	2012-01-01 04:30:00	3359.	19.0	2012-01-01	TRUE

```
# i 52,598 more rows
```

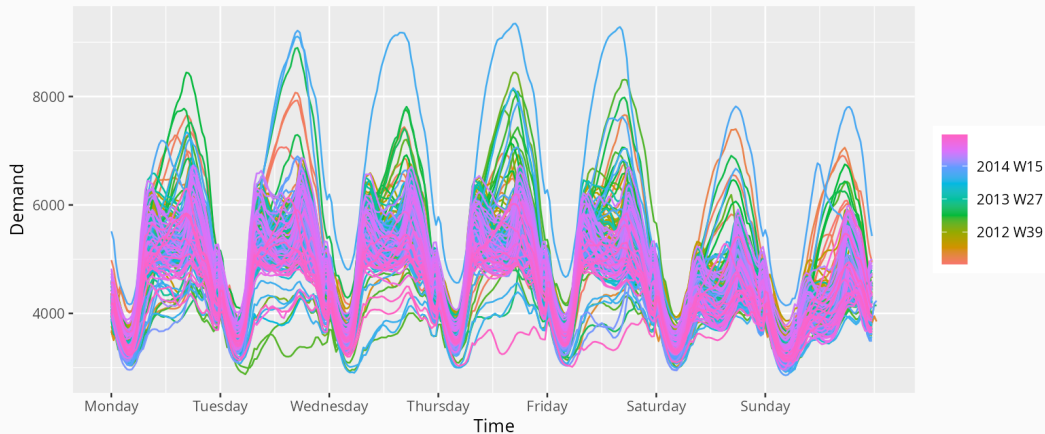
Multiple seasonal periods

```
vic_elec |> gg_season(Demand)
```



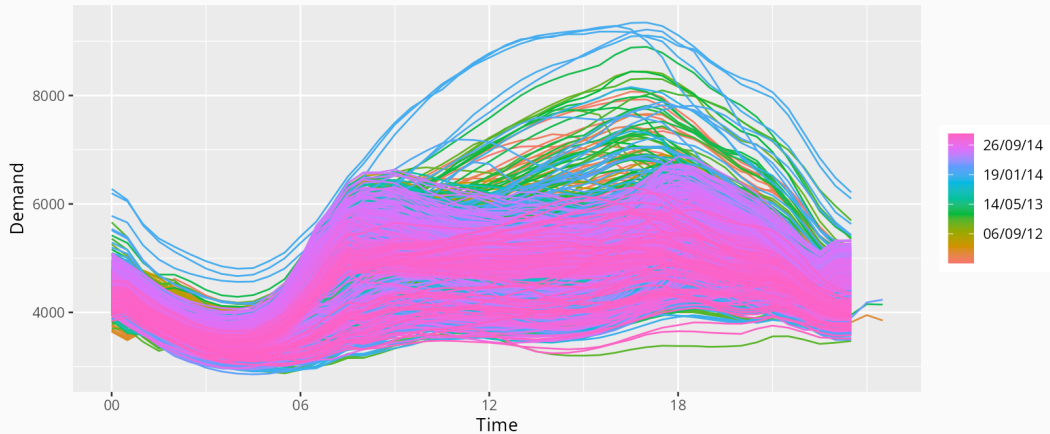
Multiple seasonal periods

```
vic_elec |> gg_season(Demand, period = "week")
```



Multiple seasonal periods

```
vic_elec |> gg_season(Demand, period = "day")
```

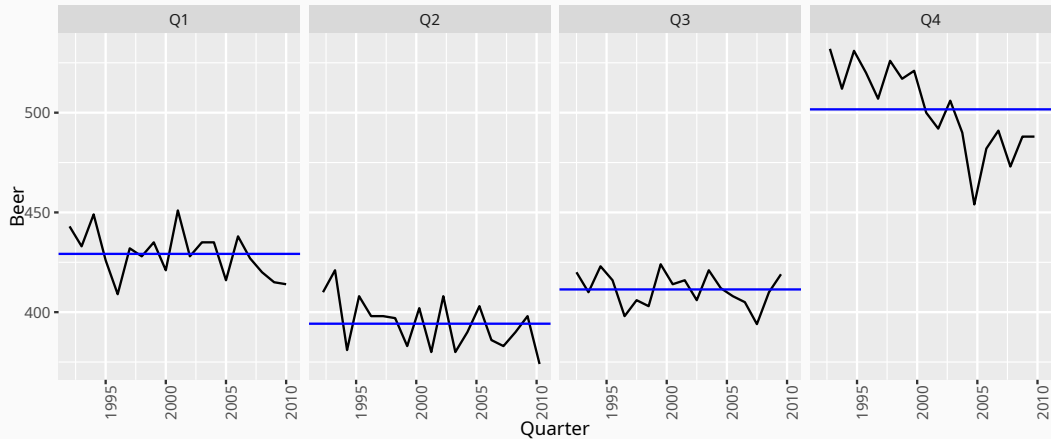


Seasonal subseries plots

- Data for each season collected together in time plot as separate time series.
- Enables the underlying seasonal pattern to be seen clearly, and changes in seasonality over time to be visualized.
- In R: `gg_subseries()` (`ggtime`)

Quarterly Australian Beer Production

```
beer |> gg_subseries(Beer)
```



Australian holidays

```
holidays <- tourism |>
  filter(Purpose == "Holiday") |>
  group_by(State) |>
  summarise(Trips = sum(Trips))
```

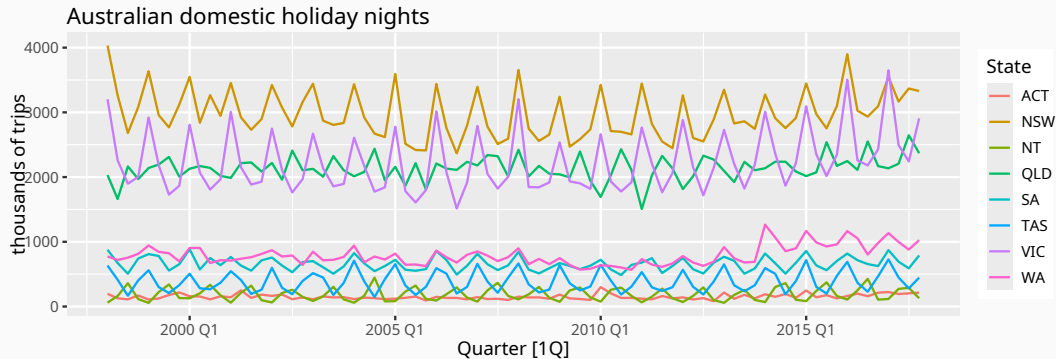
```
# A tsibble: 640 x 3 [1Q]
```

```
# Key:      State [8]
```

	State	Quarter	Trips
	<chr>	<qtr>	<dbl>
1	ACT	1998 Q1	196.
2	ACT	1998 Q2	127.
3	ACT	1998 Q3	111.
4	ACT	1998 Q4	170.
5	ACT	1999 Q1	108.
6	ACT	1999 Q2	125.
7	ACT	1999 Q3	178.
8	ACT	1999 Q4	218.
9	ACT	2000 Q1	158.
10	ACT	2000 Q2	155.

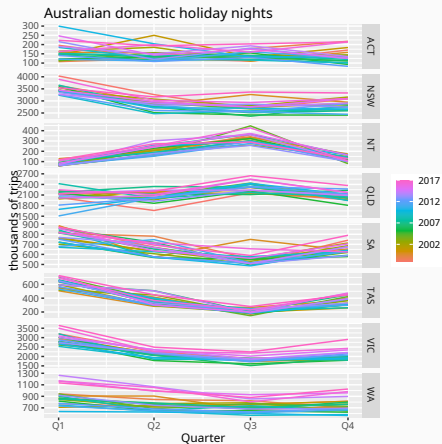
Australian holidays

```
holidays |> autoplot(Trips) +  
  labs(y = "thousands of trips", title = "Australian domestic holiday nights")
```



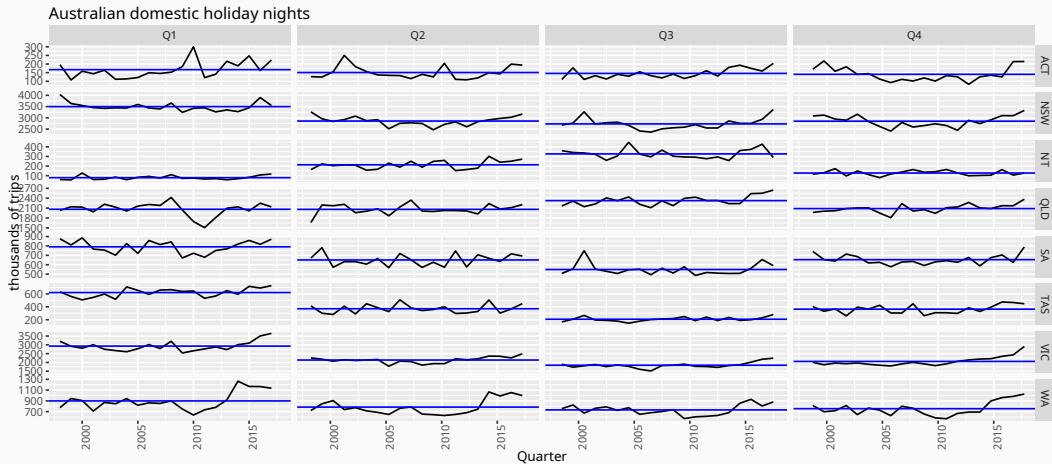
Seasonal plots

```
holidays |> gg_season(Trips) +  
  labs(y = "thousands of trips", title = "Australian domestic holiday nights")
```



Seasonal subseries plots

```
holidays |> gg_subseries(Trips) +  
  labs(y = "thousands of trips", title = "Australian domestic holiday nights")
```

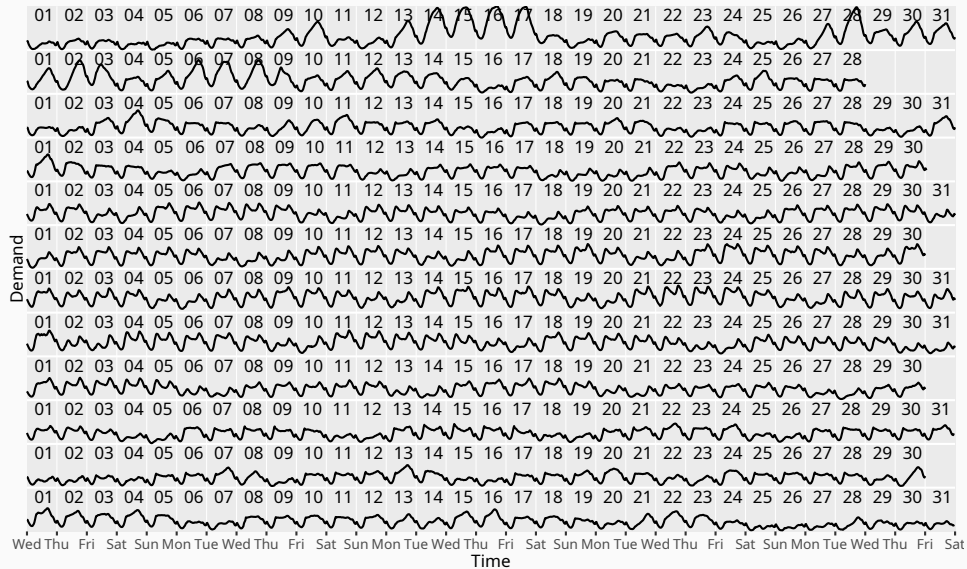


Calendar plots

```
library(ggtime)
library(mixtime)
vic_elec |>
  filter(year(Date) == year(2014L)) |>
  as_tibble() |>
  mutate(Time = datetime(Time)) |>
  ggplot(aes(x = Time, y = Demand)) +
  geom_line() +
  coord_calendar(rows = month(1L), cols = NULL) +
  scale_x_mixtime(
    time_breaks = days(1L),
    time_labels = "{cyc(day, cal_isoweek$week, label = TRUE, abbreviate = TRUE)}"
  ) +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank())
```

- `coord_calendar()` cuts a time axis into a calendar-like grid of rows/columns

Calendar plots



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Time series patterns

Trend pattern exists when there is a long-term increase or decrease in the data.

Seasonal pattern exists when a series is influenced by seasonal factors (e.g., the quarter of the year, the month, or day of the week).

Cyclic pattern exists when data exhibit rises and falls that are *not of fixed period* (duration usually of at least 2 years).

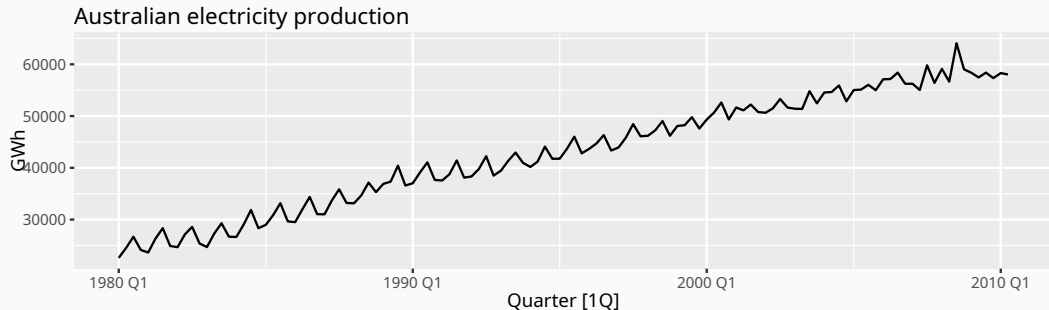
Time series components

Differences between seasonal and cyclic patterns:

- seasonal pattern constant length; cyclic pattern variable length
- average length of cycle longer than length of seasonal pattern
- magnitude of cycle more variable than magnitude of seasonal pattern

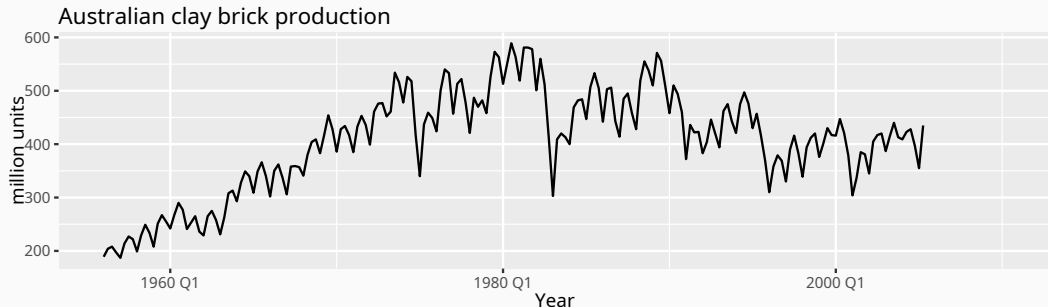
Time series patterns

```
aus_production |>  
  filter(year(Quarter) >= year(1980L)) |>  
  autoplot(Electricity) +  
  labs(y = "GWh", title = "Australian electricity production")
```



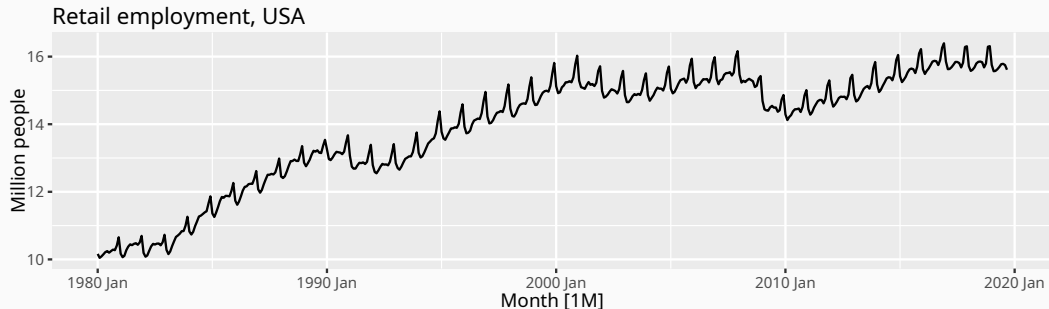
Time series patterns

```
aus_production |>  
  autoplot(Bricks) +  
  labs(title = "Australian clay brick production",  
        x = "Year", y = "million units")
```



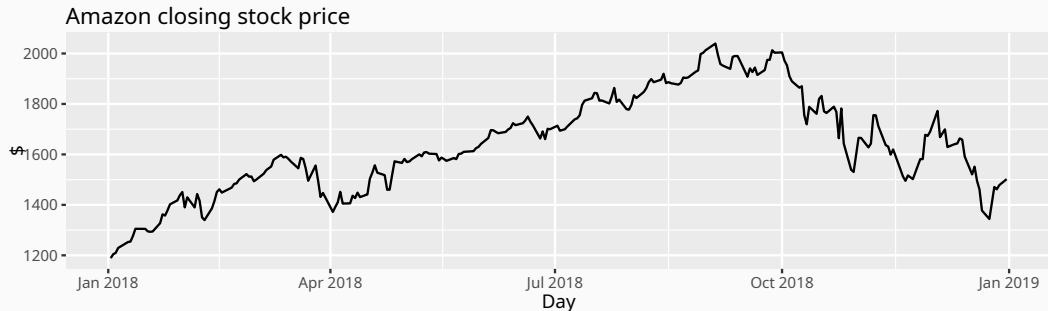
Time series patterns

```
us_employment |>  
  filter(Title == "Retail Trade", year(Month) >= year(1980L)) |>  
  autoplot(Employed / 1e3) +  
  labs(title = "Retail employment, USA", y = "Million people")
```



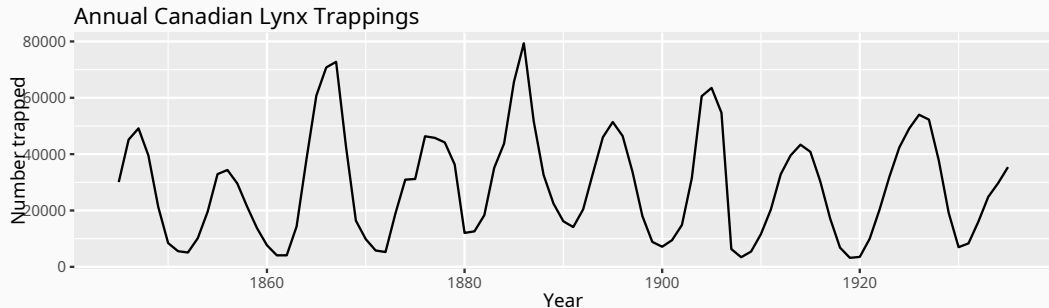
Time series patterns

```
gafa_stock |>  
  filter(Symbol == "AMZN", year(Date) >= year(2018L)) |>  
  autoplot(Close) +  
  labs(title = "Amazon closing stock price", x = "Day", y = "$")
```



Time series patterns

```
pelt |>  
  autoplot(Lynx) +  
  labs(title = "Annual Canadian Lynx Trappings",  
        x = "Year", y = "Number trapped")
```



Seasonal or cyclic?

Differences between seasonal and cyclic patterns:

- seasonal pattern constant length; cyclic pattern variable length
- average length of cycle longer than length of seasonal pattern
- magnitude of cycle more variable than magnitude of seasonal pattern

Seasonal or cyclic?

Differences between seasonal and cyclic patterns:

- seasonal pattern constant length; cyclic pattern variable length
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The timing of peaks and troughs is predictable with seasonal data, but unpredictable in the long term with cyclic data.

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Example: Beer production

```
new_production <- aus_production |>  
  filter(year(Quarter) >= year(1992L))  
new_production
```

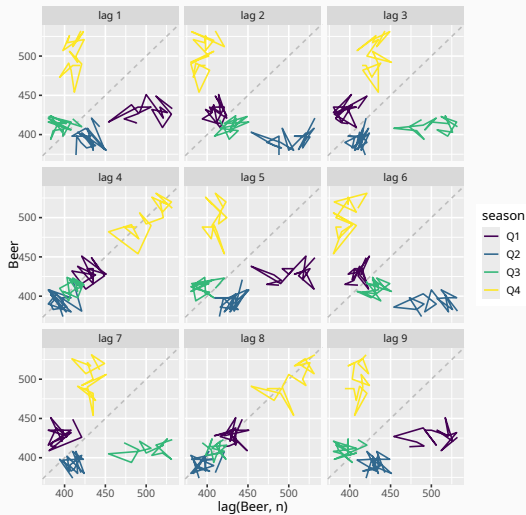
```
# A tsibble: 74 x 7 [1Q]
```

	Quarter	Beer	Tobacco	Bricks	Cement	Electricity	Gas
	<qtr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1992 Q1	443	5777	383	1289	38332	117
2	1992 Q2	410	5853	404	1501	39774	151
3	1992 Q3	420	6416	446	1539	42246	175
4	1992 Q4	532	5825	420	1568	38498	129
5	1993 Q1	433	5724	394	1450	39460	116
6	1993 Q2	421	6036	462	1668	41356	149
7	1993 Q3	410	6570	475	1648	42949	163
8	1993 Q4	512	5675	443	1863	40974	138
9	1994 Q1	449	5311	421	1468	40162	127
10	1994 Q2	381	5717	475	1755	41199	159

```
# i 64 more rows
```

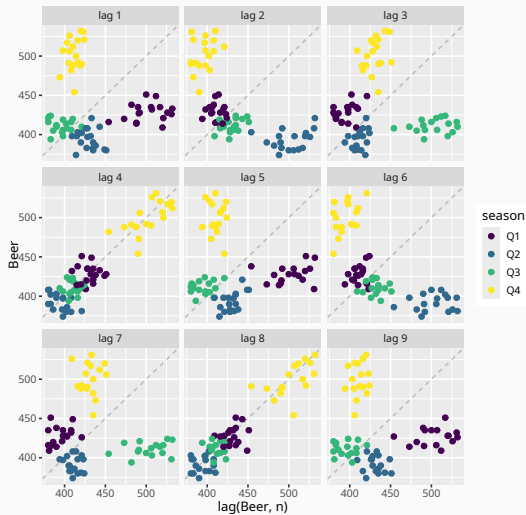
Example: Beer production

```
new_production |> gg_lag(Beer)
```



Example: Beer production

```
new_production |> gg_lag(Beer, geom = "point")
```



Lagged scatterplots

- Each graph shows y_t plotted against y_{t-k} for different values of k .
- The autocorrelations are the correlations associated with these scatterplots.
- ACF (autocorrelation function):
 - ▶ $r_1 = \text{Correlation}(y_t, y_{t-1})$
 - ▶ $r_2 = \text{Correlation}(y_t, y_{t-2})$
 - ▶ $r_3 = \text{Correlation}(y_t, y_{t-3})$
 - ▶ etc.
- If there is **seasonality**, the ACF at the seasonal lag (e.g., 12 for monthly data) will be **large and positive**.

Autocorrelation

Results for first 9 lags for beer data:

```
new_production |> ACF(Beer, lag_max = 9)
```

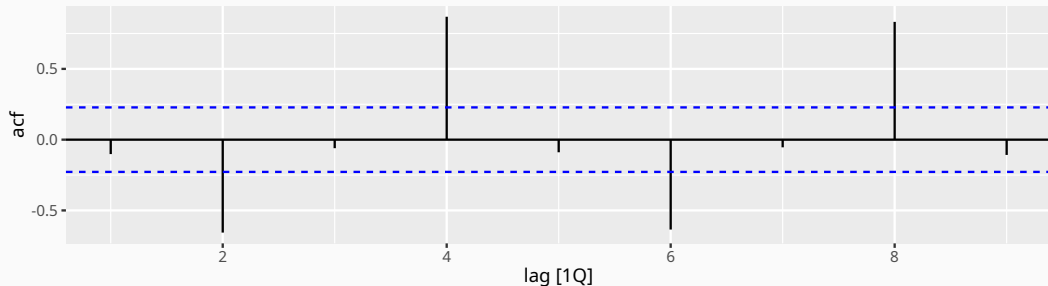
```
# A tsibble: 9 x 2 [1Q]
```

	lag	acf
	<cf_lag>	<dbl>
1	1Q	-0.102
2	2Q	-0.657
3	3Q	-0.0603
4	4Q	0.869
5	5Q	-0.0892
6	6Q	-0.635
7	7Q	-0.0542
8	8Q	0.832
9	9Q	-0.108

Autocorrelation

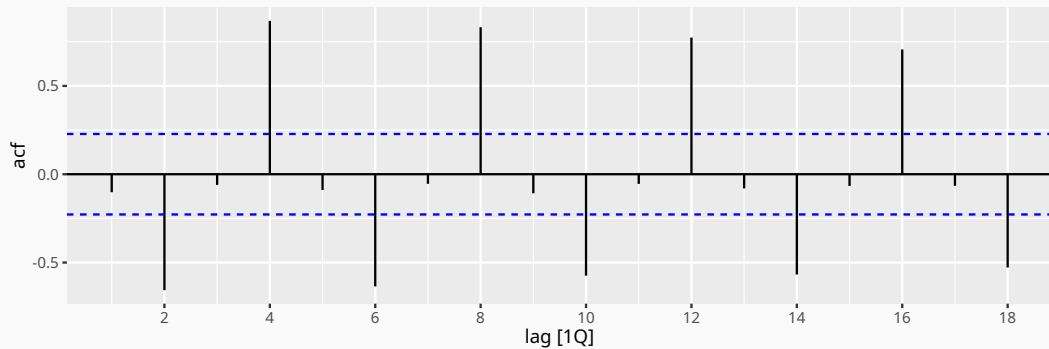
Results for first 9 lags for beer data:

```
new_production |>  
  ACF(Beer, lag_max = 9) |>  
  autoplot()
```



ACF

```
new_production |>  
  ACF(Beer) |>  
  autoplot()
```



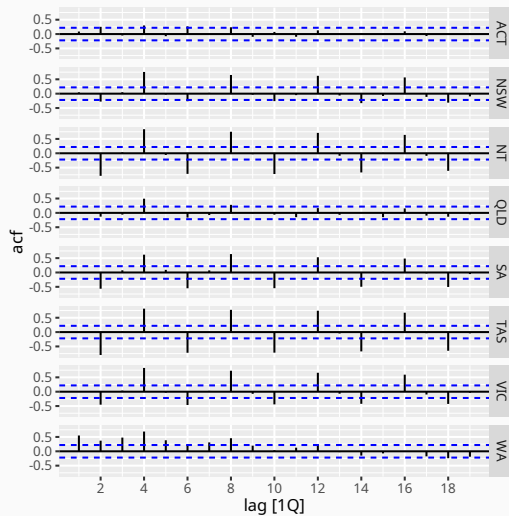
Australian holidays

```
holidays |> ACF(Trips)
```

```
# A tsibble: 152 x 3 [1Q]
# Key:      State [8]
  State      lag      acf
  <chr> <cf_lag>    <dbl>
1 ACT      1Q  0.0877
2 ACT      2Q  0.252
3 ACT      3Q -0.0496
4 ACT      4Q  0.300
5 ACT      5Q -0.0741
6 ACT      6Q  0.269
7 ACT      7Q -0.00504
8 ACT      8Q  0.236
9 ACT      9Q -0.0953
10 ACT     10Q  0.0750
# i 142 more rows
```

Australian holidays

```
holidays |> ACF(Trips) |> autoplot()
```

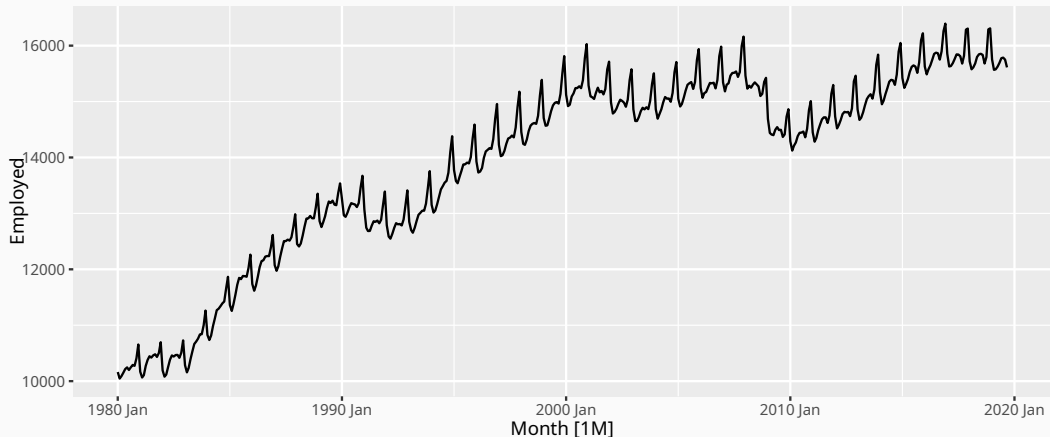


Trend and seasonality in ACF plots

- When data have a trend, the autocorrelations for small lags tend to be large and positive.
- When data are seasonal, the autocorrelations will be larger at the seasonal lags (i.e., at multiples of the seasonal frequency)
- When data are trended and seasonal, you see a combination of these effects.

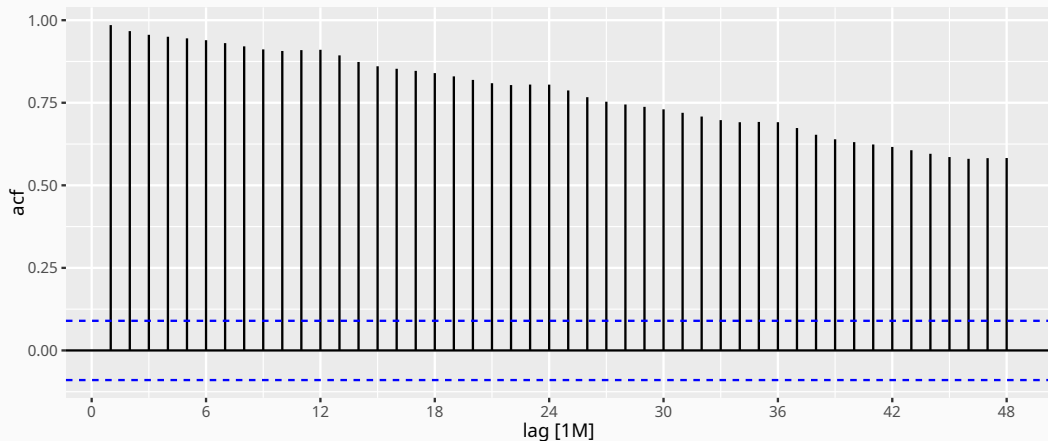
US retail trade employment

```
retail <- us_employment |>  
  filter(Title == "Retail Trade", year(Month) >= year(1980L))  
retail |> autoplot(Employed)
```



US retail trade employment

```
retail |>  
  ACF(Employed, lag_max = 48) |>  
  autoplot()
```



Google stock price

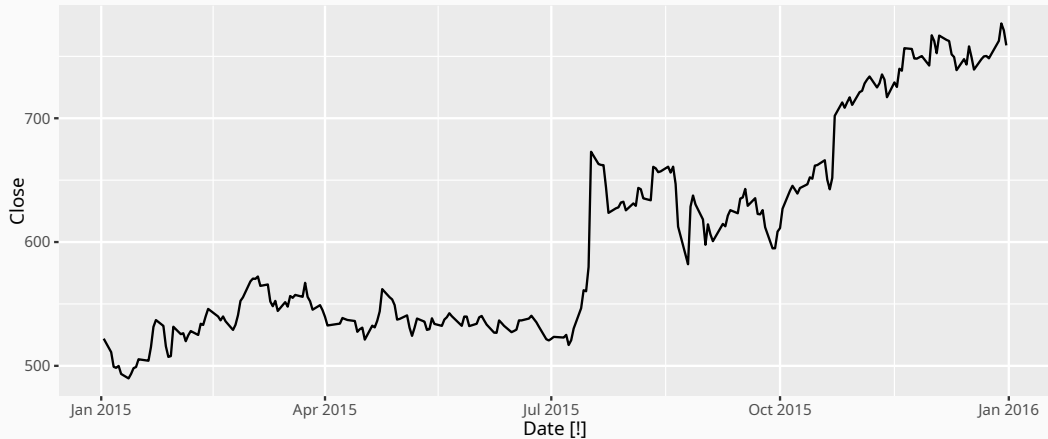
```
google_2015 <- gafa_stock |>
  filter(Symbol == "GOOG", year(Date) == year(2015L)) |>
  select(Date, Close)
google_2015
```

```
# A tsibble: 252 x 2 [!]
```

	Date	Close
	<date>	<dbl>
1	2015-01-02	522.
2	2015-01-05	511.
3	2015-01-06	499.
4	2015-01-07	498.
5	2015-01-08	500.
6	2015-01-09	493.
7	2015-01-12	490.
8	2015-01-13	493.
9	2015-01-14	498.
10	2015-01-15	499.

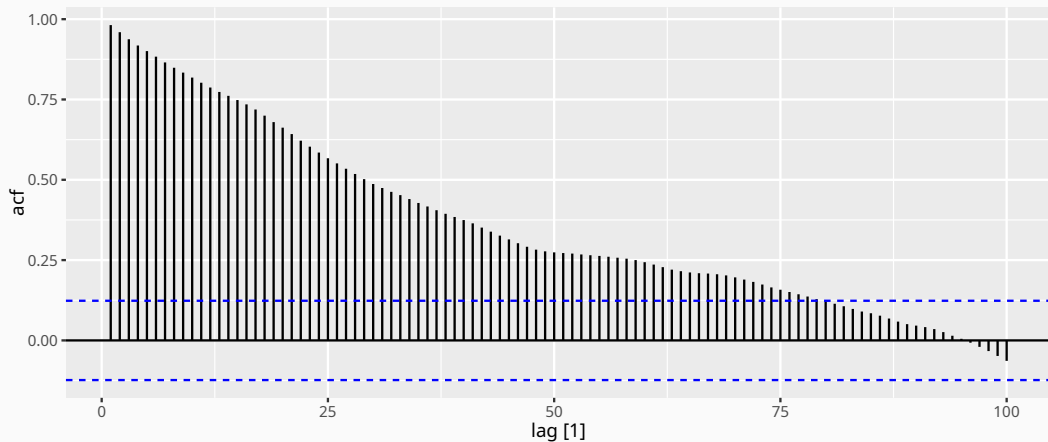
Google stock price

```
google_2015 |> autoplot(Close)
```



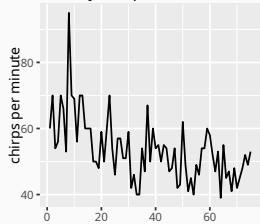
Google stock price

```
google_2015 |>  
  ACF(Close, lag_max = 100) |>  
  autoplot()
```

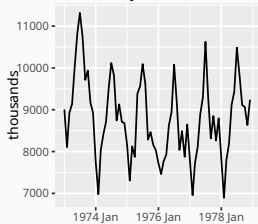


Which is which?

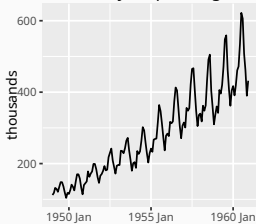
1. Daily temperature of cow



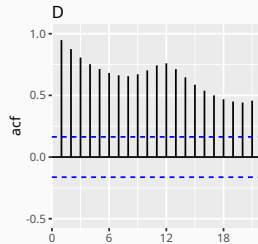
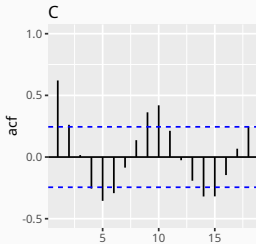
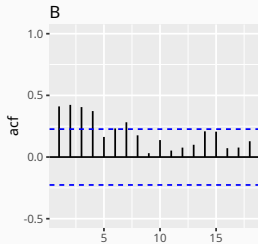
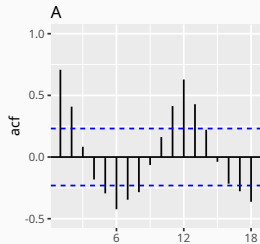
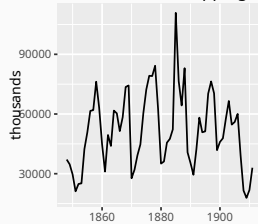
2. Monthly accidental death



3. Monthly air passengers



4. Annual mink trappings

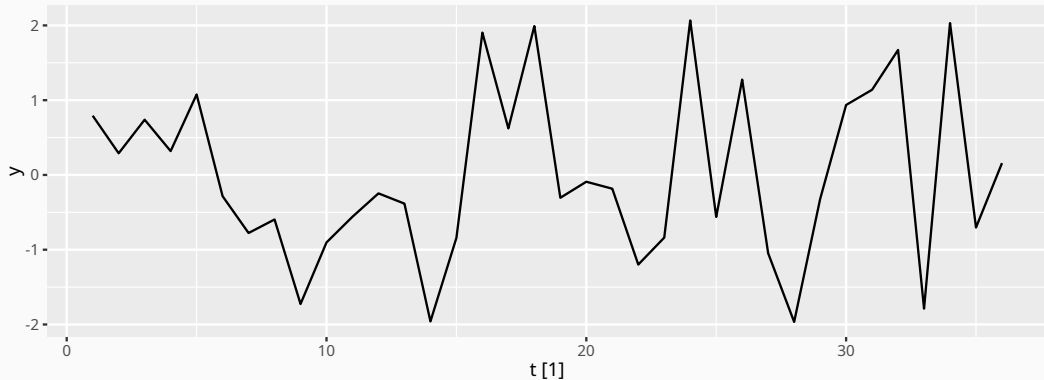


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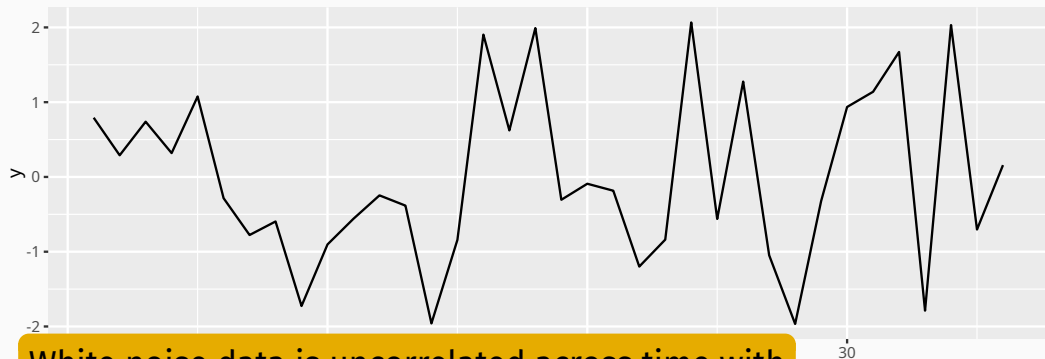
Example: White noise

```
wn <- tsibble(t = seq(36), y = rnorm(36), index = t)
wn |> autoplot(y)
```



Example: White noise

```
wn <- tsibble(t = seq(36), y = rnorm(36), index = t)
wn |> autoplot(y)
```

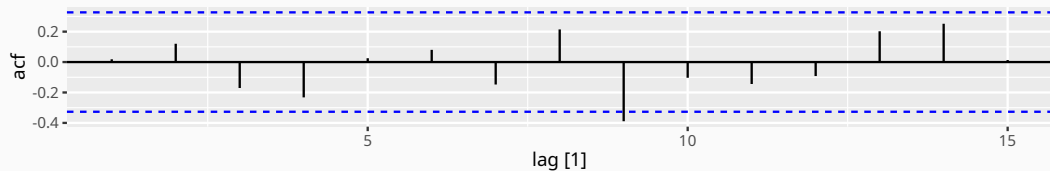


White noise data is uncorrelated across time with zero mean and constant variance.
(Technically, we require independence as well.)

Example: White noise

```
wn |> ACF(y)
```

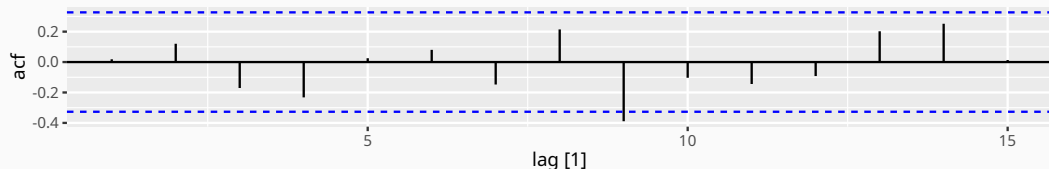
r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8	r_9	r_{10}
0.018	0.120	-0.171	-0.232	0.025	0.080	-0.148	0.215	-0.389	-0.103



Example: White noise

```
wn |> ACF(y)
```

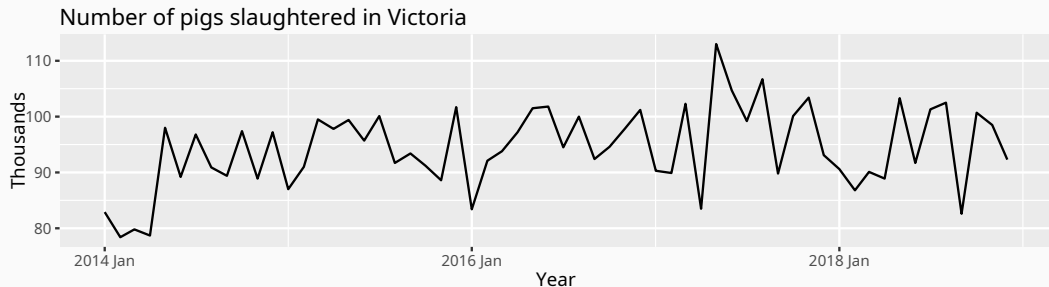
r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8	r_9	r_{10}
0.018	0.120	-0.171	-0.232	0.025	0.080	-0.148	0.215	-0.389	-0.103



- Sample autocorrelations for white noise series.
- Expect each autocorrelation to be close to zero.
- Blue lines show 95% critical values.

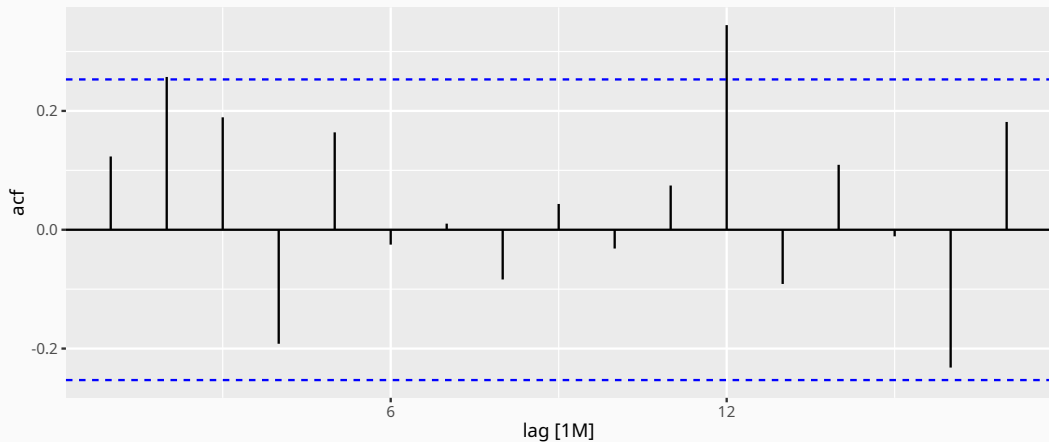
Example: Pigs slaughtered

```
pigs <- aus_livestock |>  
  filter(State == "Victoria", Animal == "Pigs", year(Month) >= year(2014L))  
pigs |> autoplot(Count / 1e3) +  
  labs(x = "Year", y = "Thousands",  
       title = "Number of pigs slaughtered in Victoria")
```



Example: Pigs slaughtered

```
pigs |>  
  ACF(Count) |>  
  autoplot()
```



Example: Pigs slaughtered

Monthly total number of pigs slaughtered in the state of Victoria, Australia, from January 2014 through December 2018
(Source: Australian Bureau of Statistics.)

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- Indicate some slight seasonality.

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These show the series is **not a white noise series**.